

David Tong Standard Model Solutions — Sheet 2: Symmetry Breaking and QCD

D. Tong, The Standard Model, Example Sheet 2; 8 worked problems.

Yucheng (Luca) Jin

Quantum Field Theory

Contents

1	Sheet 2, Problem 1: Broken Discrete Symmetry	1
2	Sheet 2, Problem 2: Matrix Order Parameters and Goldstone Modes	1
3	Sheet 2, Problem 3*: Fundamental Higgs Fields in $SU(2)$ and $SU(N)$	2
4	Sheet 2, Problem 4: Adjoint Higgs Breaking Patterns	3
5	Sheet 2, Problem 5*: One-Loop QCD Scale	3
6	Sheet 2, Problem 6: Casimirs and Dynkin Indices	4
7	Sheet 2, Problem 7: Expanding the Chiral Lagrangian	4
8	Sheet 2, Problem 8: Meson Mass Relations	5

These prompts paraphrase David Tong's The Standard Model, Example Sheet 2 (February 2026). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 2, Problem 1: Broken Discrete Symmetry

Construct a polynomial potential for a complex scalar ϕ whose $\mathbb{Z}_N$ symmetry breaks spontaneously. Identify every ground state.

Solution. The potential

$$V(\phi) = \lambda|\phi|^{2N} - \frac{1}{2}\mu^2(\phi^N + \phi^{*N}) \quad (1.1)$$

is invariant under $\mathbb{Z}_N : \phi \rightarrow e^{2\pi i k/N} \phi$. Therefore, the vacuum is given by

$$v_k = \left(\frac{\mu^2}{2\lambda}\right)^{1/N} e^{2\pi i k/N}. \quad (1.2)$$

2. Sheet 2, Problem 2: Matrix Order Parameters and Goldstone Modes

Let $M(x)$ be a complex $N \times N$ matrix with $\lambda > 0$ and

$$S = \int d^4x \operatorname{Tr} \left[\partial^\mu M^\dagger \partial_\mu M - k M^\dagger M - \frac{\lambda}{2} M^\dagger M M^\dagger M \right]. \quad (2.1)$$

- Show invariance under $M \mapsto A M B^\dagger$ for $A, B \in U(N)$ and identify the $U(1)$ subgroup of $U(N) \times U(N)$ that acts trivially.
- When $k < 0$, show that a vacuum satisfies $M_0^\dagger M_0 = v^2 \mathbf{1}$. Find the unbroken group, describe the vacuum manifold as a coset, and count Goldstone bosons.
- Add $\int d^4x h(\det M + \det M^\dagger)$. Determine the new symmetry group and the Goldstone count if the vacuum still has $M_0^\dagger M_0 = v^2 \mathbf{1}$ with $v \neq 0$.

Solution.

- The invariance is trivial to see. For $A = B \in U(1)$, the transformation is trivial.

ii) The differentiation of the potential with respect to $M^\dagger$, we find

$$\frac{\partial}{\partial M^\dagger} \left(\frac{\lambda}{2} M_0^\dagger M_0 M_0^\dagger M_0 + k M_0^\dagger M_0 \right) = M_0 (\lambda M_0^\dagger M_0 + k) = 0 \implies M_0^\dagger M_0 = -\frac{k}{\lambda} \mathbf{1}. \quad (2.2)$$

Therefore, the vacuum must take the form $M_0 = vU$ with $U \in U(N)$. We can choose $U = \mathbf{1}$. Under $U(N) \times U(N)$, the vacuum transforms as

$$v\mathbf{1} \rightarrow Av\mathbf{1}B^\dagger. \quad (2.3)$$

Therefore, the vacuum is invariant for $A = B \in U(N)$. Therefore, the stability group is given by $H = U(N)$, while the coset space is $G = (U(N) \times U(N))/U(N)$. Therefore, the number of Goldstone bosons is $\dim G = N^2$.

iii) With this additional term, the symmetry reduces to $A, B \in SU(N)$, namely $SU(N) \times SU(N)$. If the vacuum still takes the same form, then the stability group becomes $H = SU(N)$. Hence, the number of Goldstone bosons becomes $\dim SU(N) = N^2 - 1$.

3. Sheet 2, Problem 3*: Fundamental Higgs Fields in $SU(2)$ and $SU(N)$

For a fundamental $SU(2)$ scalar ϕ_a ($a = 1, 2$), use

$$S = \int d^4x \left[-\frac{1}{2} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + D_\mu \phi^\dagger D^\mu \phi - \frac{\lambda}{2} (\phi^\dagger \phi - v^2)^2 \right], \quad (3.1)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$, $D_\mu \phi = \partial_\mu \phi - igA_\mu \phi$, and $T^a = \sigma^a/2$ in the fundamental representation. Find the particle masses. Generalise to a fundamental scalar in $SU(N)$: after condensation, what symmetry survives, how many gauge bosons become massive, and how many real scalar components are eaten?

Solution. For $SU(2)$, we can choose the vacuum to be

$$\phi_0 = \begin{pmatrix} 0 \\ v \end{pmatrix}. \quad (3.2)$$

Therefore, all of the symmetries are broken, and we can reparameterise the scalar field as

$$\phi(x) = \begin{pmatrix} 0 \\ v + \sigma(x) \end{pmatrix} e^{i\pi^a(x)T^a/F_\pi}, \quad (3.3)$$

where F_π is chosen to be v so that the kinetic term of π is canonically normalised. The gauge boson acquires a mass through the term

$$g^2 A_\mu A^\mu |\phi|^2 = v^2 g^2 A_\mu A^\mu + \dots, \quad (3.4)$$

where $m_A = \sqrt{2}vg$. The $\pi(x)$ field is eaten by the gauge bosons. The mass of $\sigma(x)$ is given by

$$\lambda v^2 |\phi|^2 = \lambda v^2 \sigma^2 + \dots, \quad (3.5)$$

which gives $m_\sigma = \sqrt{2\lambda}v$.

For $SU(N)$, a choice of vacuum could be

$$\phi_0 = \begin{pmatrix} 0 \\ \vdots \\ v \end{pmatrix}. \quad (3.6)$$

Therefore, the stability group is $H = SU(N-1)$, and hence the coset space is $G = SU(N)/SU(N-1)$. Therefore, we can reparameterise the scalar field as

$$\phi(x) = m \begin{pmatrix} 0 \\ \vdots \\ v + \sigma(x) \end{pmatrix} e^{i\pi^a(x)T^a/F_\pi}, \quad (3.7)$$

where T^a are the generators of G and $a = 1, \dots, \dim G$. For consistency, $\phi(x)$ has $2N$ degrees of freedom, while $\pi(x)$ has $\dim G = 2N - 1$ degrees of freedom and $\sigma(x)$ has 1 degree of freedom. All of the π are eaten by the gauge fields. Therefore, there will be $2N - 1$ massive bosons.

4. Sheet 2, Problem 4: Adjoint Higgs Breaking Patterns

An $SU(N)$ adjoint scalar is a traceless matrix $\phi = \phi^A T^A$ with $D_\mu \phi = \partial_\mu \phi - ig[A_\mu, \phi]$. Explain why a potential minimum can be represented by

$$\phi_0 = \text{diag}(v_1, \dots, v_N), \quad \sum_{a=1}^N v_a = 0, \quad v_a \leq v_{a+1}. \quad (4.1)$$

Classify the residual gauge groups according to coincidences among the v_a .

Solution. Given a vacuum ϕ_0 , we can always diagonalise it by $U\phi_0U^\dagger$, where $U \in SU(N)$. Therefore, we can choose the vacuum to be

$$\phi_0 = \text{diag}(v_1, \dots, v_N), \quad v_a \leq v_{a+1}. \quad (4.2)$$

Since T^A are traceless, we must have $\text{Tr} \phi_0 = 0$, which implies $\sum_a v_a = 0$. For a gauge transformation generated by T^A to be unbroken, we must have

$$[T^A, \phi_0] = 0. \quad (4.3)$$

In general, we have

$$[T^A, \phi_0]_{ik} = \sum_j T_{ij}^A \delta_{jk} v_k - \sum_j \delta_{ij} v_j T_{jk}^A = (v_k - v_i) T_{ik}^A. \quad (4.4)$$

Therefore, there are no restrictions on T_{ii} , which gives $N - 1$ unbroken generators. Meanwhile, we must either have $T_{ik} = 0$ for all $i \neq k$, or $v_i - v_k = 0$ for some i, k . Therefore, the number of unbroken generator is equal to the number of identical eigenvalue pairs. Suppose each distinct eigenvalues have multiplicity n_i , they give

$$2 \sum_j \binom{n_j}{2} = \sum_j (n_j^2 - n_j) = \sum_j n_j^2 - N. \quad (4.5)$$

Therefore, the total number of unbroken generators is

$$\sum_j n_j^2 - 1. \quad (4.6)$$

5. Sheet 2, Problem 5*: One-Loop QCD Scale

For $SU(N_c)$ with N_f massless fundamental Dirac fermions, use

$$\frac{1}{g^2(\mu)} = \frac{1}{g_0^2} - \frac{11N_c - 2N_f}{3(4\pi)^2} \log \frac{\Lambda_{\text{UV}}^2}{\mu^2}. \quad (5.1)$$

Taking $\alpha_s = g_s^2/(4\pi) \simeq 0.12$ at $\mu = M_Z \simeq 90 \text{ GeV}$, estimate Λ_{QCD} if every quark lighter than M_Z is treated as massless. How can the estimate be made more realistic?

Solution. Taking the physical scale, we have

$$\frac{1}{g^2(M_Z)} = \frac{1}{g_0^2} - \frac{11N_c - 2N_f}{3(4\pi)^2} \log \frac{\Lambda_{\text{UV}}^2}{M_Z^2}, \quad (5.2)$$

while the Landau pole is at

$$0 = \frac{1}{g^2(\Lambda_{\text{QCD}})} = \frac{1}{g_0^2} - \frac{11N_c - 2N_f}{3(4\pi)^2} \log \frac{\Lambda_{\text{UV}}^2}{\Lambda_{\text{QCD}}^2}. \quad (5.3)$$

Combining the two, we find

$$\frac{11N_c - 2N_f}{3(4\pi)^2} \log \frac{M_Z^2}{\Lambda_{\text{QCD}}^2} = \frac{1}{g^2(M_Z)} \implies \Lambda_{\text{QCD}} = M_Z \exp \left(-\frac{6\pi}{(11N_c - 2N_f)\alpha_s} \right). \quad (5.4)$$

Five quark flavours are lighter than M_Z : u, d, s, c, b . The top quark is heavier, so take $N_c = 3$ and $N_f = 5$. This gives $\Lambda_{\text{QCD}} \approx 0.097 \text{ GeV}$.

For a more realistic estimation, we can run the coupling towards the IR. When we go across the boundary of the bottom quark mass, we take N_f to be 4. Then, we match the value at $\alpha(m_b)$ in both regime. We repeat this procedure until ran into the landau pole.

6. Sheet 2, Problem 6: Casimirs and Dynkin Indices

For a representation R , define

$$T^A T^A = C(R) \mathbf{1}, \quad \text{Tr}(T^A T^B) = \frac{1}{2} I(R) \delta^{AB}. \quad (6.1)$$

Prove $2C(R) \dim(R) = I(R) \dim(G)$. Find $C(R)$ for the fundamental and anti-fundamental of $SU(N)$, and calculate the adjoint Casimir of $SU(2)$.

Solution.

$$\text{Tr}(T^A T^A) = C(R) \dim \mathbf{1} = C(R) \dim R, \quad (6.2)$$

while

$$\text{Tr}(T^A T^A) = \frac{1}{2} I(R) \delta^{AA} = \frac{1}{2} I(R) \dim G. \quad (6.3)$$

Therefore,

$$2C(R) \dim(R) = I(R) \dim(G). \quad (6.4)$$

For fundamental representation of $SU(N)$, we have $\dim G = \dim \mathbf{N}$ and $I(\mathbf{N}) = 1$, and hence $C(\mathbf{N}) = \frac{1}{2}$. Since $\bar{T}^A = -(T^A)^*$, we must have $I(\mathbf{N}) = I(\bar{\mathbf{N}})$, and hence $C(\mathbf{N}) = C(\bar{\mathbf{N}}) = 1/2$.

For adjoint representation of $SU(2)$, the generators are $-i\epsilon^i_{jk}$, and hence

$$i\epsilon^i_{jk} i\epsilon^i_{kl} = 2\delta_{jl} \implies C(\text{adj}) = 2. \quad (6.5)$$

7. Sheet 2, Problem 7: Expanding the Chiral Lagrangian

For $U = e^{2i\pi/f_\pi}$ with $\pi \in su(N_f)$, expand

$$\mathcal{L}_{\text{pion}} = \frac{f_\pi^2}{4} \text{tr}(\partial^\mu U^\dagger \partial_\mu U) \quad (7.1)$$

through fourth order in π . Show that the quadratic term is $\text{tr}(\partial_\mu \pi)^2$ and that the quartic term can be written

$$-\frac{2}{3f_\pi^2} \text{tr} [\pi^2 (\partial_\mu \pi)^2 - (\pi \partial_\mu \pi)^2]. \quad (7.2)$$

For $N_f = 2$ and $T^a = \sigma^a/2$, reduce the interaction to

$$\mathcal{L}_{\text{int}} = -\frac{1}{6f_\pi^2} [\pi^a \pi^a \partial_\mu \pi^b \partial^\mu \pi^b - \pi^a \partial_\mu \pi^a \pi^b \partial^\mu \pi^b]. \quad (7.3)$$

Solution. Expanding U , we find

$$U = 1 + \frac{2i}{f_\pi}\pi - \frac{2}{f_\pi^2}\pi^2 - \frac{4i}{3f_\pi^3}\pi^3 + \dots, \quad (7.4)$$

and hence

$$\partial_\mu U = \frac{2i}{f_\pi}\partial_\mu\pi - \frac{2}{f_\pi^2}((\partial_\mu\pi)\pi + \pi(\partial_\mu\pi)) - \frac{4i}{3f_\pi^3}((\partial_\mu\pi)\pi^2 + \pi(\partial_\mu\pi)\pi + \pi^2(\partial_\mu\pi)) + \dots. \quad (7.5)$$

Expanding the Lagrangian, we have

$$\begin{aligned} \mathcal{L}_{\text{pion}} &= \text{tr}(\partial_\mu\pi\partial^\mu\pi) - \frac{2}{3f_\pi^2}\text{tr}[\pi^2(\partial_\mu\pi)^2 - (\pi\partial_\mu\pi)^2] + \dots. \end{aligned} \quad (7.6)$$

For $N_f = 2$, we need to evaluate

$$\text{tr}(\sigma^a\sigma^b\sigma^c\sigma^d) = \text{tr}(\sigma^a\sigma^b\{\sigma^c, \sigma^d\}) - \text{tr}(\sigma^a\{\sigma^b, \sigma^d\}\sigma^c) + \text{tr}(\{\sigma^a, \sigma^d\}\sigma^b\sigma^c) + \text{tr}(\sigma^d\sigma^a\sigma^b\sigma^c). \quad (7.7)$$

Using $\{\sigma^a, \sigma^b\} = 2\delta^{ab}$ and $\text{Tr}\{\sigma^a\sigma^b\} = 2\delta^{ab}$, we find

$$\text{tr}(\sigma^a\sigma^b\sigma^c\sigma^d) = 2(\delta^{ab}\delta^{cd} - \delta^{bd}\delta^{ac} + \delta^{ad}\delta^{bc}). \quad (7.8)$$

Substituting into the Lagrangian, the interaction reduces to

$$\mathcal{L}_{\text{int}} = -\frac{1}{6f_\pi^2}[\pi^a\pi^a\partial_\mu\pi^b\partial^\mu\pi^b - \pi^a\partial_\mu\pi^a\pi^b\partial^\mu\pi^b]. \quad (7.9)$$

8. Sheet 2, Problem 8: Meson Mass Relations

For $N_f = 3$, use the pion, kaon, and eta matrix

$$\pi = \frac{1}{\sqrt{2}} \begin{pmatrix} \pi^0/\sqrt{2} + \eta/\sqrt{6} & \pi^+ & K^+ \\ \pi^- & -\pi^0/\sqrt{2} + \eta/\sqrt{6} & K^0 \\ K^- & \bar{K}^0 & -2\eta/\sqrt{6} \end{pmatrix}. \quad (8.1)$$

With $M = \text{diag}(m_u, m_d, m_s)$ and a mass term proportional to $f_\pi \text{tr}[(M + M^\dagger)\pi^2]$, show

$$\frac{m_{K^+}^2 - m_{K^0}^2}{m_\pi^2} = \frac{m_u - m_d}{m_u + m_d}. \quad (8.2)$$

Then take $m_u \simeq m_d$ to derive the Gell-Mann–Okubo relation $4m_K^2 \simeq 3m_\eta^2 + m_\pi^2$ and compare it with measured meson masses.

Solution.

$$\begin{aligned} (\pi^2)_{11} &= \frac{1}{2} \left[\left(\frac{\pi^0}{\sqrt{2}} + \frac{\eta}{\sqrt{6}} \right)^2 + \pi^+\pi^- + K^+K^- \right], \\ (\pi^2)_{22} &= \frac{1}{2} \left[\left(-\frac{\pi^0}{\sqrt{2}} + \frac{\eta}{\sqrt{6}} \right)^2 + \pi^-\pi^+ + K^0\bar{K}^0 \right], \\ (\pi^2)_{33} &= \frac{1}{2} \left[K^-K^+ + \bar{K}^0K^0 + \frac{2\eta^2}{3} \right]. \end{aligned}$$

Hence, the mass term is proportional to

$$m_u(\pi^2)_{11} + m_d(\pi^2)_{22} + m_s(\pi^2)_{33}. \quad (8.3)$$

Then, we can read off the masses of the Mesons, which are

$$m_\pi^2 = m_u + m_d, \quad m_{K^0}^2 = m_d + m_s, \quad m_{K^+}^2 = m_u + m_s, \quad m_\eta^2 = \frac{m_u + m_d + 4m_s}{3}, \quad (8.4)$$

up to an overall constant. Therefore, we obtain the relation

$$\frac{m_{K^+}^2 - m_{K^0}^2}{m_\pi^2} = \frac{m_u - m_d}{m_u + m_d}. \quad (8.5)$$

Take $m_u \simeq m_d$, we immediately find $m_K^2 \simeq m_{K^0}^2 \simeq m_{K^+}^2 \simeq m_u + m_s$. Then, it follows

$$4m_K^2 \simeq 3m_\eta^2 + m_\pi^2. \quad (8.6)$$

Using the PDG measured masses and averaging the charged and neutral squared masses within each pion and kaon multiplet gives

$$\begin{aligned} 4m_K^2 &\simeq 0.983 \text{ GeV}^2. \\ 3m_\eta^2 + m_\pi^2 &\simeq 0.920 \text{ GeV}^2. \end{aligned}$$